SUNGKYUN CHANG

PhD Student, Centre for Digital Music (C4DM), Queen Mary University of London

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SUMMARY

Music AI researcher working on generative modelling and transcription of music audio. Developed YourMT3+ (multi-instrument transcription) and SpanSynth-Edit (MIDI-guided audio synthesis and editing); co-author of ImprovNet, Text2Score, and CMI-RewardBench. Interested in agentic music production and RL on top of transcription and generative models. Trained composer (B.M.) with a background in audio DSP.

EDUCATION

Queen Mary University of London *Sep 2026 – present*

PhD in Computer Science, London, UK

– Centre for Digital Music (C4DM). Supervisors: Prof. Simon Dixon, Dr. Emmanouil Benetos. Fully funded by an ABRSM industrial PhD studentship.

Seoul National University *2010 – 2013*

M.S. in Engineering, Digital Contents Convergence, Seoul, Korea

– Graduate School of Convergence Science and Technology. Thesis: *Modeling tonal tension in music signals*. Advisor: Prof. Kyogu Lee.

Seoul National University *1999 – 2008*

B.M. in Music Composition, Seoul, Korea

– College of Music.

EXPERIENCE

Research Assistant (Doctoral Researcher since Sep 2026) *Jun 2024 – present*

Centre for Digital Music, Queen Mary University of London · ABRSM collaboration (PI: Prof. Simon Dixon), London, UK

– SpanSynth-Edit: MIDI-guided audio synthesis and editing of multi-instrument mixtures using scalar-quantised latents with MIDI span conditioning (under review, ICASSP 2027).

– Co-authored ImprovNet (IJCNN 2025), a corruption–refinement model for controllable symbolic improvisation; Text2Score (text-to-sheet-music generation); and CMI-RewardBench (ICML 2026), a benchmark for music reward models.

– Music performance assessment and feedback: score-to-performance alignment, transcription, and mistake detection (RUMAA, WASPAA 2025); a pilot study of LLM-based assessment and feedback understanding for piano exams (DMRN+20, 2025); towards a unified model that predicts exam marks and generates text feedback from MusicXML scores, audio, and text.

Research Assistant *Jul 2022 – Dec 2023*

Centre for Digital Music, Queen Mary University of London · Huawei collaboration (PI: Dr. Emmanouil Benetos), London, UK

– Developed YourMT3+, a multi-instrument automatic music transcription model: Mixture-of-Experts and Perceiver-based encoder, multi-stream decoder for partially annotated data, and cross-dataset stem augmentation; improved results on 10 public datasets and pop music recordings (MLSP 2024).

– Released it as an open-source PyTorch toolkit for multi-task, multi-dataset training (GitHub / Hugging Face demo).

Research Scientist *Apr 2019 – Jun 2022*

Cochl.ai, Seoul, Korea

– Neural audio fingerprinting for large-scale music retrieval using self-supervised contrastive learning (ICASSP 2021); deployed in SK Telecom FLO's music identification service, where it replaced Gracenote and remains in production.
– Self-supervised and meta-learning for sound event detection.

Researcher *Jun 2017 – Mar 2019*
Institute for Industrial System Innovation, Seoul National University · Collaboration with Kakao and Kakao Brain, Seoul, Korea

– Deep learning for cover song identification (NeurIPS 2017 ML4Audio Workshop; MIREX 2018 winner; ICASSP 2018), music genre classification (WWW 2018 challenge), and playlist skip prediction (WSDM Cup 2019, top 3).
– Contributed to an end-to-end singing synthesis system built on WaveNet and Tacotron 2, with speech-to-singing transfer through a hierarchical latent space.

Graduate Researcher *2010 – 2017*
Music and Audio Research Group (MARG), Seoul National University, Seoul, Korea

– Unsupervised lyrics-to-audio alignment (IEEE Access 2017), singing voice onset/offset detection (ICASSP 2014), and music descriptor extraction for Naver; projects funded by NRF, Samsung, and MSIT.

SELECTED PUBLICATIONS

[13] **Synthesis and editing of multi-instrument audio mixtures using scalar-quantised latents with MIDI span conditioning**
S. Chang, K. Bhandari, S. Dixon, E. Benetos
Under review, ICASSP 2027 · Poster at Audio Developer Conference (ADC) 2026, online

[12] **Text2Score: Generating sheet music from textual prompts**
K. Bhandari, S. Chang, A. Roy, F. Ronchini, E. Benetos, D. Herremans, S. Colton
Under review, AAAI 2027

[11] **CMI-RewardBench: Evaluating music reward models with compositional multimodal instruction**
Y. Ma, H. Xia, H. Gao, W. Chen, Y. Ye, Y. Yang, S. Chang, M. Ding, Y. Li, et al.
ICML 2026

[10] **LLM-based assessment and feedback understanding for music performance exams**
H. Zhang, S. Chang, E. Benetos, S. Dixon
DMRN+20: Digital Music Research Network One-day Workshop, 2025

[9] **RUMAA: Repeat-aware unified music audio analysis for score-performance alignment, transcription, and mistake detection**
S. Chang, S. Dixon, E. Benetos
IEEE WASPAA 2025

[8] **ImprovNet: Generating controllable musical improvisations with iterative corruption refinement**
K. Bhandari, S. Chang, T. Lu, F. Enus, L. B. Bradshaw, D. Herremans, S. Colton
IJCNN 2025

[7] **YourMT3+: Multi-instrument music transcription with enhanced transformer architectures and cross-dataset stem augmentation**
S. Chang, E. Benetos, H. Kirchhoff, S. Dixon
IEEE MLSP 2024

[6] **Neural audio fingerprint for high-specific audio retrieval based on contrastive learning**
S. Chang, D. Lee, J. Park, H. Lim, K. Lee, K. Ko, Y. Han
IEEE ICASSP 2021

[5] **Sequential skip prediction with few-shot in streamed music contents**
S. Chang, S. Lee, K. Lee
WSDM Cup 2019 Workshop

[4] **Audio cover song identification using convolutional neural network**
S. Chang, J. Lee, S. K. Choe, K. Lee
NeurIPS 2017 Workshop on Machine Learning for Audio

[3] **Cover song identification using song-to-song cross-similarity matrix with convolutional neural network**
J. Lee, S. Chang, S. K. Choe, K. Lee
IEEE ICASSP 2018

[2] **Lyrics-to-audio alignment by unsupervised discovery of repetitive patterns in vowel acoustics**
S. Chang, K. Lee
IEEE Access, vol. 5

[1] **A pairwise approach to simultaneous onset/offset detection for singing voice using correntropy**
S. Chang, K. Lee
IEEE ICASSP 2014

Also: OmniVideoBench (ICLR 2026), 2025 AMT Challenge report (NeurIPS 2025 AI for Music Workshop), MixPlore (Int. J. Arts and Technology, 2011). Full list on Google Scholar.

GRANTS & FELLOWSHIPS

Fully Funded PhD Studentship (tuition and stipend), Associated Board of the Royal Schools of Music (ABRSM) — *2026*

EuroHPC JU GPU Grant, European Union: 35,000 GH200 GPU-hours for *AMIE-Scale: Developing a Scalable Transcription-Complete Music Audio-Language Model* (research lead; PIs: E. Benetos and S. Dixon) — *2026*

TPU Research Cloud Grant, Google — *2022*

HONORS & AWARDS

Winner, Music Tech Hackathon, Music Hackspace, London — *2025*

Winner, AES AIMLA Query-by-Vocal-Imitation Challenge — *2025*

2nd Place, IEEE ICME Automatic Music Transcription Challenge — *2025*

Top 3, WSDM Cup 2019: Spotify Sequential Skip Prediction Challenge — *2019*

Winner, MIREX 2018: Audio Cover Song Identification — *2018*

Finalist, WWW 2018 Challenge: Learning to Recognize Musical Genre — *2018*

PATENTS

Method and apparatus for generating music fingerprints (KR 10-2020-0113961; SK Telecom & Cochl.ai) — *2021*

Apparatus and method for script and scene alignment for multimedia sorting, analysing, and tagging (KR 10-2014-0017363; Seoul National University) — *2014*

Real-time musical performance feedback system for beginner musicians (KR 2015-0034959; Seoul National University) — *2013*

ACADEMIC SERVICE

REVIEWER ICASSP (2022–), IEEE/ACM TASLP (2023–), ISMIR (2021–), IJCNN (2023–), TISMIR (2025–), Journal of New Music Research (2023–), IEEE ICME (2025–), IEEE Signal Processing Letters (2022–), IEEE OJSP (2026–), IEEE Access (2018–)

COMMITTEE Program Committee, AAAI Workshop on Emerging AI Technologies for Music (EAIM) (2025–)